Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of eco-friendly automated processes for producing highly functional silicone products**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	*Organic/Inorganic Material & catalyst manufacturing technologies*	*Polymeric Materials*	*Polymerization Reactions / Process Technologies*
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The main methods of this project is to develop a CF (Continuous Flow) synthesis for BARCs and RDL materials (batch and flow), controlled addition of water and acid/base catalyst under varying solvent conditions using green chemistry principles in order to drive the synthesis in a reliable and sustainable way. Some selected principles of Green Chemistry (https://www.compoundchem.com/2015/09/24/green-chemistry/) will be considered, in particular: • Waste prevention (principle 1): • Less hazardous chemical synthesis (principle 3): • Designing safer chemicals (principle 4): • Safer solvents and auxiliaries (principle 5): • Design for energy efficiency (principle 6 • Safer chemistry for accident prevention (principle 12):			

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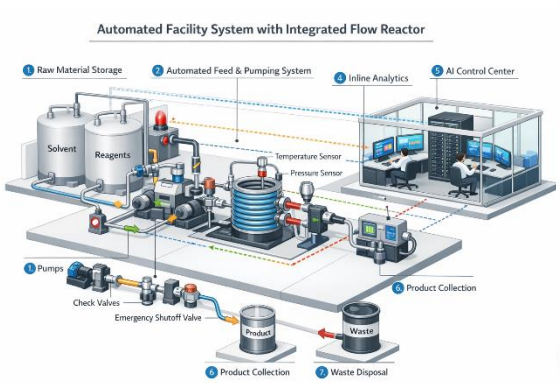
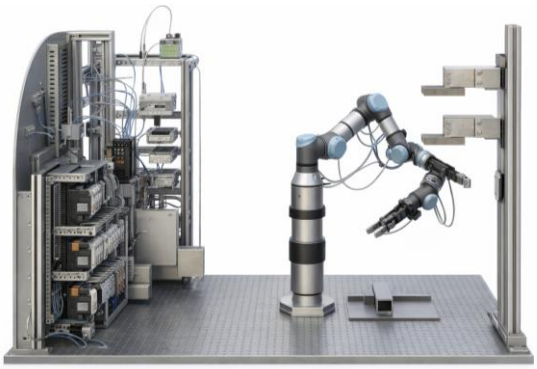
	Furthermore, material synthesis & formulation, lithographic evaluation, metallization process, and analytical testing validation will be part of the aims and methods.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The transition from batch synthesis to CF synthesis is challenging since chemical reactions are strongly affected. However, a CF synthesis promises a more robust, easy scalable and controllable synthesis. Therefore, the project will enable the project partner to increase quality and quantity of siloxane-based materials for customer. There will be siloxane materials that have high-functional properties and show high performance. An automated facility system including flow reactor will enable the company to have extended inline control during synthesis. Using AI methods, the synthesis system can also be featured by optimization routines.   **Necessity**: The design and synthesis of high-performance materials such as BARCs and RDL for semiconductor applications requires a multidisciplinary approach that integrates expertise in materials chemistry, process engineering, and device testing. While local capabilities may provide a strong foundation in polymer synthesis or flow chemistry, access to cutting-edge research is required. **Objective:** The jointly design and validate flow-based synthesis protocols for advance semiconductors, enabling reproducible, scalable production suitable for integration energy material pipelines and have to explore device-level testing and application integration of siloxane polymers.
Key Words	1. Highly functional silicone / siloxane products 2. Automated and robotic flow synthesis 3. Eco-friendly chemical synthesis.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• Development of highly functional siloxane materials • Development of environmentally friendly and automated process synthesis • Development of sustainable chemical synthesis: techniques and methodologies			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* • Develop new synthesize processes for to-date BARC formulations aiming for synthesis amount of > 1 l / day • Construct meso-reactor setup for flow-based hydrolysis-condensation of siloxane polymers including in-line sensing/analysis methods and adaption of synthesis control, keeping variations below 3 % • Deliver lithographic performance report of BARC reducing reflection to < 2 % • Formulate, photolithographic patterning, complete electrical testing of RDL and validated performance.			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* 1. Definition of high-functional and high-performance siloxane based monomers, oligomers, and polymers for the application of semiconductor or display industry, synthesize by conventional batch synthesis: Essential material properties remeasured as benchmark 2. Second step will be Design and development of automated reactor system including flow reactor system. CF Syntheses products will be compared with batch products 3. In line measurement tools will be integrated in order to get inline information of synthesis quality 4. AI based algorithms will adapt synthesis conditions when in-line data are not within specified threshold 5. Materials will be tested as BARC materials and RDL material 6. An Assessment about Green Chemistry principles will be carried out to prove sustainability.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* • Determination of variables and parameters to optimize or stabilize syntheses (temperature, pH concentration, opaqueness.) for siloxane based oligomers and/or polymers for envisioned applications • Evaluation of online sensing /measuring of individual parameters to stabilize/optimize syntheses • Adaptation of synthesis to principles of Green Chemistry • Development of Machine Learning algorithm to steer synthesis conditions based on the online measurement; Development of automated synthesis systems • Consultation to integrate measurement devices and controlling procedures into syntheses processes at Industrial partner
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* • Production and commercialization of siloxane materials applicable to semiconductor, display, and rechargeable battery • Joint design and development of automated reactor system and set up and operate in-house system • Acquire joint ownership rights of patents and exclusive license for commercialization of materials, processes, and systems

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* The outcome of the project will be that the Korean company will have knowledge about built up of automated CF synthesis processes for siloxane- based materials that are to be used for semiconductor and display industry. The quality and reproducibility of materials will be strongly enhanced. In the same time, sustainability is improved by following Green Chemistry principles. The automatization of the synthesis ends up in cheaper production free of errors of operators. The AI approach will support that synthesis conditions are always stable and led to reproducible synthesis products.

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization